# Interpretable Machine Learning

# Conditional Feature Importance (CFI)

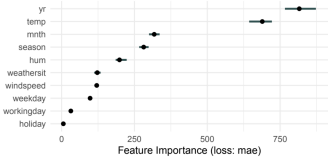
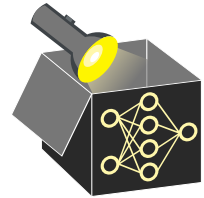



**Figure:** Bike Sharing Dataset

## Learning goals

- Extrapolation and Conditional Sampling
- Conditional Feature Importance (CFI)
- Interpretation of CFI and difference to PFI

# CONDITIONAL FEATURE IMPORTANCE IDEA



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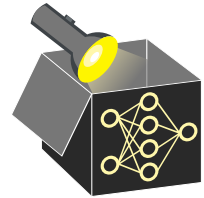

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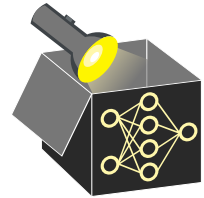

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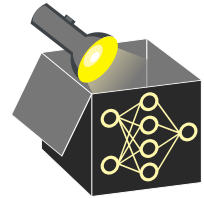

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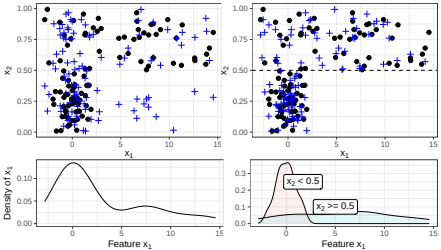
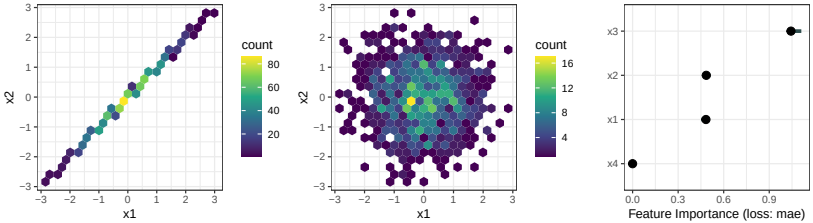
**Example:** Conditional permutation scheme ▸ Molnar et. al (2020)




- $X_2 \sim U(0,1)$ and $X_1 \sim N(0,1)$ if $X_2 < 0.5$, else $X_1 \sim N(4,4)$ (black dots)
- **Left:** For $X_2 < 0.5$, permuting $X_1$ (crosses) preserves marginal (but not joint) distribution
  $\rightsquigarrow$ Bottom: Marginal density of $X_1$
- **Right:** Permuting $X_1$ within subgroups $X_2 < 0.5$ & $X_2 \geq 0.5$ reduces extrapolation
  $\rightsquigarrow$ Bottom: Density of $X_1$ conditional on groups

# RECALL: EXTRAPOLATION IN PFI

**Example:** Let $y = x_3 + \epsilon_y$ with $\epsilon_y \sim N(0, 0.1)$ where $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$ are highly correlated ($\epsilon_1 \sim N(0,1), \epsilon_2 \sim N(0, 0.01)$) and $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim N(0,1)$. All noise terms are independent. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.



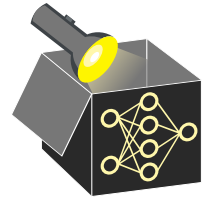

Hexbin plot of $x_1, x_2$ before permuting $x_1$ (left), after permuting $x_1$ (center), and PFI scores (right) $\Rightarrow$ $x_1$ and $x_2$ should be irrelevant for the prediction $\hat{f}(\mathbf{x})$ for

$\{\mathbf{x} : \mathbb{P}(\mathbf{x}) > 0\}$ as $0.3x_1 - 0.3x_2 \approx 0$
$\Rightarrow$ PFI evaluates model on unrealistic obs. outside $\mathbb{P}(\mathbf{x})$ $\rightsquigarrow$ $x_1$ and $x_2$ are considered relevant

# CONDITIONAL FEATURE IMPORTANCE

Strobl et al. (2008)

Hooker et al. (2021)



Conditional feature importance (CFI) for features $x_S$ using test data $\mathcal{D}$:

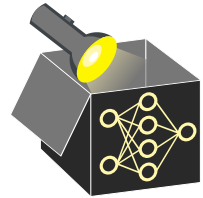
- Measure the error **with unperturbed features**.
- Measure the error **with perturbed feature values** $\tilde{x}^{S|-S}$, where $\tilde{x}_S^{S|-S} \sim \mathbb{P}(x_S|x_{-S})$
- Repeat permuting the feature (e.g., $m$ times) and average the difference of both errors:

$$\widehat{CFI}_S = \frac{1}{m}\sum_{k=1}^{m} \mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^{S|-S}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

Here, $\tilde{\mathcal{D}}^{S|-S}$ denotes the dataset where features $x_S$ where sampled conditional on the remaining features $x_{-S}$.

# IMPLICATIONS OF CFI König et al. (2020)

**Interpretation:** Due to the conditional sampling w.r.t. all other features, CFI quantifies a feature's unique contribution to the model performance.



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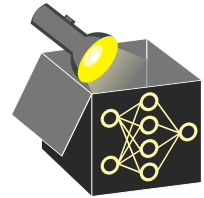

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**Entanglement with data:**

- If feature $x_S$ does not contribute unique information about $y$, i.e., $x_S \perp\!\!\!\perp y | x_{-S} \Rightarrow$ CFI $= 0$
- Why? Under the conditional independence $\mathbb{P}(\tilde{x}^{S|-S}, y) = \mathbb{P}(x, y)$ $\rightsquigarrow$ no prediction-relevant information is destroyed by permutation of $x_S$ conditional on $x_{-S}$

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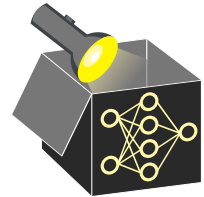

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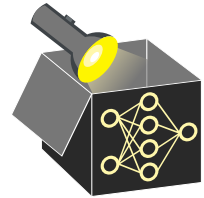
**Entanglement with model:**

- If the model does not use a feature $\Rightarrow$ CFI $= 0$
- Why? Then the prediction is not affected by any perturbation of the feature
  $\rightsquigarrow$ model performance does not change after conditional permutation

# IMPLICATIONS OF CFI



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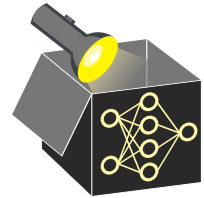

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   - If $x_j \not\perp\!\!\!\perp y$ but $x_j \perp\!\!\!\perp y | x_{-j}$ (e.g., $x_j$ and $x_{-j}$ share information) $\Rightarrow CFI_j = 0$
   - $x_j$ is not exploited by model (regardless of whether it is useful for $y$ or not) $\Rightarrow CFI_j = 0$

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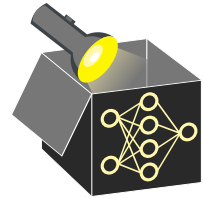

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3. Does the model require access to $x_j$ to achieve its prediction performance?
   - $CFI_j \neq 0 \Rightarrow x_j$ contributes unique information (meaning $x_j \not\perp\!\!\!\perp y | x_{-j}$)
   - Only uncovers the relationships that were exploited by the model

# COMPARISON: PFI AND CFI

**Example:** Let $y = x_3 + \epsilon_y$ with $\epsilon_Y \sim N(0, 0.1)$ where $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$ are highly correlated ($\epsilon_1 \sim N(0,1), \epsilon_2 \sim N(0, 0.01)$) and $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim N(0,1)$. All noise terms are independent. Fitting a LM yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$.

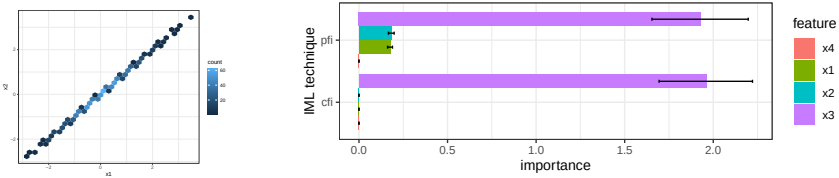


**Figure:** Density plot for $x_1, x_2$ before permuting $x_1$ (left). PFI and CFI (right).

$\Rightarrow$ $x_1$ and $x_2$ are irrelevant for the prediction $\hat{f}(\mathbf{x})$ for $\{\mathbf{x} : \mathbb{P}(\mathbf{x}) > 0\}$ as $0.3x_1 - 0.3x_2 \approx 0$

$\Rightarrow$ PFI evaluates model on unrealistic obs. outside $\mathbb{P}(\mathbf{x})$ $\rightsquigarrow$ $x_1, x_2$ are considered relevant (PFI $> 0$)

$\Rightarrow$ Since $x_1$ can be reconstructed from $x_2$ and vice versa, CFI considers $x_1$ and $x_2$ to be irrelevant